

Asra Nasir

AI & Machine Learning Researcher
Lahore, Pakistan

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PROFESSIONAL SUMMARY

AI and Machine Learning researcher with an MS in Computer Science and 4 published papers in deep learning, computer vision, and federated learning. Experienced in developing multimodal recognition systems, medical imaging models, and agricultural AI solutions. Currently teaching undergraduate CS courses as visiting faculty at two institutions.

EDUCATION

MS Computer Science Kinnaird College for Women, Lahore Aug 2023 — Jun 2025	CGPA 3.04
BS Computer Science The University of Lahore Sep 2019 — Aug 2023	CGPA 3.06

EXPERIENCE

Institute of Arts & Culture / University of Education Lecturer — Visiting Faculty <i>CS Instruction · Course Design · Student Mentoring</i> - Deliver lectures and design course materials, assignments, and assessments for undergraduate computer science programs across two institutions. - Conduct interactive sessions and mentor students on academic projects and research.	Sep 2025 — Present
Kinnaird College for Women Graduate Researcher — MS Thesis <i>Python · TensorFlow · PyTorch · OpenCV · Deep Learning · Computer Vision</i> - Thesis: Developed a multimodal Pakistani Urdu Sign Language recognition and emotion detection system using deep learning. - Built CNN-based models for Pakistani currency detection and classification for visually impaired individuals; published in IEEE ICoDT.	Aug 2023 — Jun 2025
WAPDA — IT Department Internee IT <i>MySQL · Web Maintenance · Government IT Operations</i> Maintained official WAPDA websites and managed employee databases using MySQL; collaborated with teams in a government setting.	Aug — Oct 2022
The University of Lahore Undergraduate Researcher <i>Flutter · Machine Learning · Python</i> - Developed "PakRiders" ride-sharing mobile app using Flutter as final-year capstone project. - Built ML models for diabetic outcome prediction and PSL 2023 match prediction using AI/ML techniques.	Sep 2019 — Aug 2023

PUBLICATIONS

[1] Nasir, A. et al. "A Transfer Learning-Based VGG-16 Framework for Automated Pneumonia Detection from Chest X-Ray Images with Real-Time Web Deployment" — SES, vol. 4(2), pp. 516–537, Feb 2026

[2] Nasir, A. et al. "A Novel Approach for Pakistani Urdu Sign Language Alphabets Recognition and Classification using Various Deep Learning Models" — Data Intelligence, Sep 2025

[3] Nasir, A. et al. "A Novel Maize Plant Disease Classification using Federated Machine Learning in Smart Agriculture" — Data Intelligence

[4] Nasir, A. et al. "Pakistani Currency Recognition and Classification for Visually Impaired People Using Convolutional Neural Network" — IEEE ICoDT, pp. 1–6, Oct 2024

TECHNICAL SKILLS

Languages	Python, SQL, Dart (Flutter)
AI & ML	TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Deep Learning, Computer Vision, Federated Learning
Tools	Jupyter, Google Colab, WordPress, Advanced Excel, Mendeley, LaTeX
Soft Skills	Teaching & Mentoring, Research Writing, Team Collaboration, Leadership

CREDENTIALS & HONORS

Cambridge Direct Examinations — Invigilation & Supervision Beaconhouse Defence Campus · Jun 2024
Mendeley in Research Cycle — Training HEC Pakistan · Nov 2023
IEEE Xplore — Literature Review Training IEEE · Oct 2023
Ad Mad Competition Winner Kinnaird Business Week · Mar 2022
The Citizens Foundation — Summer Camp Internship Jul 2019

REFERENCES

Dr. Sidra Zafar — Assistant Professor, Kinnaird College for Women, Lahore | sidra.zafar@kinnaird.edu.pk

Dr. Ghousia Usman — Assistant Professor, Kinnaird College for Women, Lahore | ghousia.usman@kinnaird.edu.pk

Ms. Arfa Dilawari — Lecturer, Kinnaird College for Women, Lahore | arfa.dilawari@kinnaird.edu.pk